

UNIT II

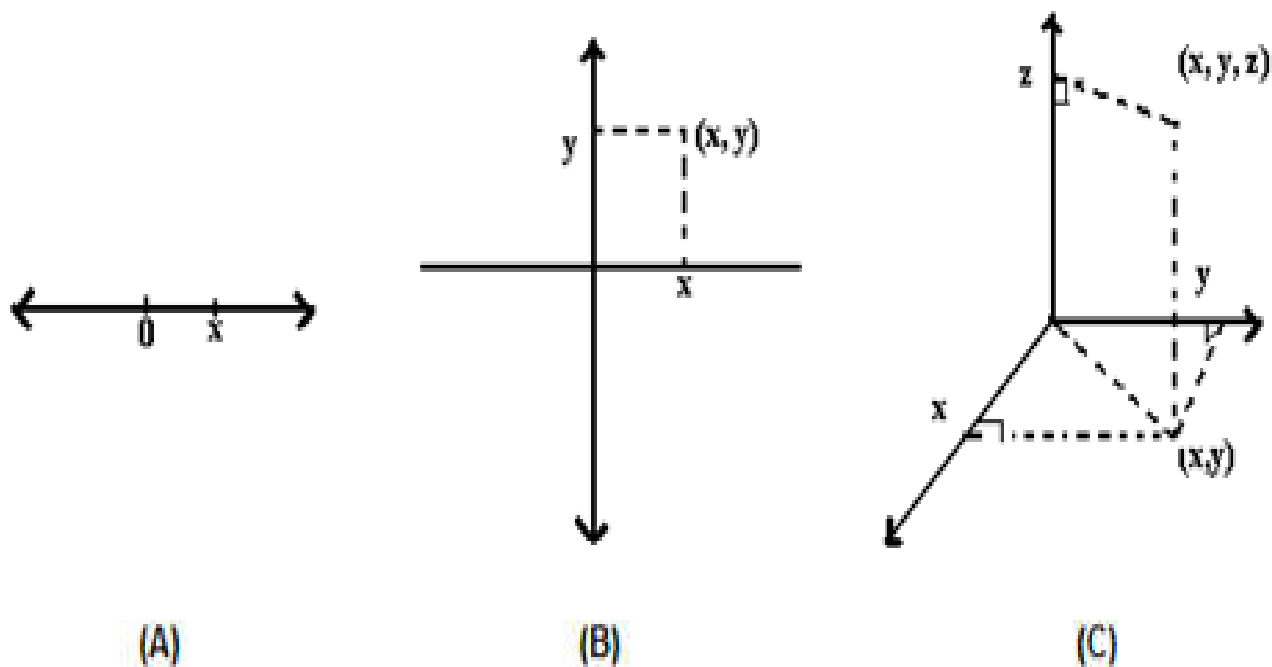
VECTORS

Certain Physical quantities such as mass, area, density, volume, etc., that possess only magnitude are called *scalars*. On the other hand, there are physical quantities such as force, displacement, velocity, acceleration, etc that has both magnitude and direction. Such quantities are called *vectors*.

The concept of a vector is essential for the whole course. It provides the foundation and geometric motivation for everything that follows. Hence the properties of vectors, both algebraic and geometric, will be discussed in this unit.

2.1 Definition of Points in n-space

We know that, once a unit length is selected, a number x can be used to represent a point on a line. A pair of numbers (i.e. a couple of numbers) (x, y) can be used to represent a point in the plane. A triple of numbers (x, y, z) can be used to represent a point in space. The following pictures illustrate these representations:



We can say that a single number represents a point in 1-space (A), a couple represents a point in 2-space (B) and a triple represents a point in 3-space (C).

Although we cannot draw a picture to go further, we can say that a quadruple of numbers (x, y, z, w) or (x_1, x_2, x_3, x_4) represent a point in 4-space.

Activity 2.1.1: Define a point in n -space, where n is a positive integer.

The set of all points in n -space is represented by $\mathbb{R}^n$. For the point X in $\mathbb{R}^n$, represented by n -tuple of real numbers $(x_1, x_2, \dots, x_n)$, the numbers $x_1, x_2, \dots, x_n$ are called the **coordinates** of X .

If $A = (a_1, a_2, \dots, a_n)$ and $B = (b_1, b_2, \dots, b_n)$ are points in the same space $\mathbb{R}^n$, and if c is a real number then

- i. A and B are equal (or represent the same point) if $a_1 = b_1, a_2 = b_2, \dots$ and $a_n = b_n$.
- ii. $A + B, A - B$ and cA are defined to be the points whose coordinates are $(a_1 + b_1, a_2 + b_2, \dots, a_n + b_n), (a_1 - b_1, a_2 - b_2, \dots, a_n - b_n)$ and $(ca_1, ca_2, \dots, ca_n)$, respectively.

Example 2.1.2 1) Let $A = (1, 2), B = (-3, 4)$, then $A+B=(-2, 6), A-B = (4, -2), -3A=(-3, -6)$
2) Let $X = (1, 0, \pi, 4), Y = (2, 4, -2\pi, -6)$, then $2X+Y = (4, 4, 0, 2)$ and $X-(1/2) Y = (0, -2, 2\pi, 7)$.

2.2. Vectors in n-space, Geometric interpretation in 2 and 3- spaces

Every pair of distinct points A and B in $\mathbb{R}^n$ determines a directed line segment with initial point at A and terminal point at B. We call such a directed line segment a **vector** and denote it by $\overrightarrow{AB}$. The length of the line segment is the magnitude of the vector.

Although $\overrightarrow{AA}$ has zero length, and strictly speaking, no direction, it is convenient to view it as a vector. It is called a **zero** or a **null vector**. It is often denoted by $\vec{0}$.

Two vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ will be considered to be equal (or equivalence), $\overrightarrow{AB} = \overrightarrow{CD}$, if they have the same magnitude and direction.

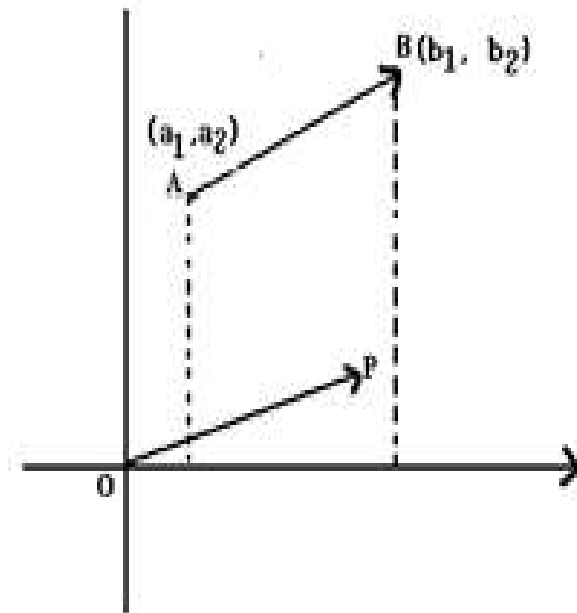
Notice that the definition of equality of two vectors does not require that the vectors have the same initial and terminal points. Rather it suggests that we can move vectors freely provided we make no change in magnitude and direction.

Activity 2.2.1

Let (a_1, a_2) be the coordinate representation of A and let (b_1, b_2) be that of B . Let P be the point $(b_1 - a_1, b_2 - a_2)$. If O is the origin, is $\vec{OP}$ in the direction of $\vec{AB}$?

Is length of $\vec{OP}$ equal to the length of $\vec{AB}$?

Is $\vec{OP} = \vec{AB}$?



If your answer for the above questions is yes, then we can conclude that any vector $V = \overrightarrow{AB}$ in the plane is a vector $\vec{OP}$ with initial point at the origin. This is the only vector whose initial point is the origin and $P = B - A$, which is equal to $\overrightarrow{AB}$. Moreover, $V = \vec{OP}$ is uniquely determined by its terminal point P . If $P = (x, y)$, then we shall write $V = (x, y)$ and refer to it as the coordinate representation of V relative to the chosen coordinate system. In view of this, we shall call (x, y) either a point or a vector, depending on the interpretation which we have in mind. So if $V = \overrightarrow{AB}$, then we can write $V = B - A$. In view of this two vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are equal (or equivalence) if $B - A = D - C$.

Example 1.2.1: If $P = (1, 3)$, $Q = (-1, 0)$, $R = (0, -1)$ and $S = (-2, -4)$ then $\vec{QP} = \vec{RS}$

As numbers can be added subtracted and multiplies, vectors can be combined in the following ways.

Let $A = (a_1, a_2)$ and $B = (b_1, b_2)$ vectors and t be a real number. The sum

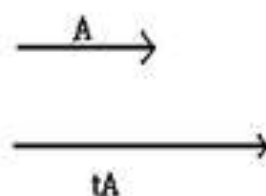
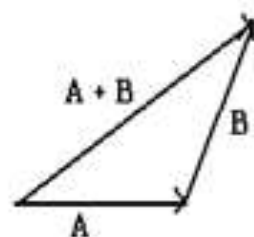
$$A + B = (a_1 + b_1, a_2 + b_2)$$

The difference $A - B = (a_1 - b_1, a_2 - b_2)$

The scalar multiple $tA = (t a_1, t a_2)$

The geometric interpretation of the above vector operations is that $A + B$ is a vector obtained by placing the initial point of B on the terminal point of A .

If $t > 0$, then tA is a vector in the direction of A . What about if $t < 0$? A and tA are said to have opposite direction.(see figure a and b)



for $t > 0$ Fig. a



for $t < 0$ Fig. b

We can extend the above notions to vectors in $\mathbb{R}^n$ but the geometric interpretations for $n > 3$ are difficult. Hence we focus on algebraic aspects of vectors.

If $A = (a_1, a_2, \dots, a_n)$ and $B = (b_1, b_2, \dots, b_n)$ are vectors in $\mathbb{R}^n$ and if t is any real number, then

$$A + B = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$tA = (ta_1, ta_2, \dots, ta_n)$$

A and B are said to be **parallel** vectors iff either $A = tB$ for some real number t or $B = tA$. In other words, A and B are parallel iff one is a scalar multiple of the other.

Using the arrow notation $\vec{PQ}$ and $\vec{RS}$ are parallel if and only if

$$\vec{Q} - \vec{P} = t(\vec{S} - \vec{R}) \text{ or } \vec{S} - \vec{R} = t(\vec{Q} - \vec{P}) \text{ for some real number } t.$$

Activity 2.2.2: 1) Let $A = (6, -2, 4)$. Find two vectors C and D which are parallel to A .

Are C and D also parallel to each other?

2) Let $P = (3, 7)$, $Q = (-4, 2)$, $R = (5, 1)$, $S = (-16, -14)$.

Is $\vec{PQ}$ parallel to $\vec{RS}$?

Using the above definitions and applying the associative and commutative properties of real numbers, one can prove the following theorem.

Theorem 2.2.1 Let A , B and C be any members of $\mathfrak{R}^n$, and let m and n be any real numbers. Then

- | | | |
|--------------------------------|---|-----------------------|
| a) $m(nA) = (mn)A$ | } | Associativity |
| b) $(A + B) + C = A + (B + C)$ | | |
| c) $A + B = B + A$ | | <u>Commutativity</u> |
| d) $(m + n)A = mA + nA$ | } | Distributive property |
| e) $m(A + B) = mA + mB$ | | |
| f) $0A = O$ | | |
| g) $A + O = A$ | | |

Proof c) $A + B = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)$

$$= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$= (b_1 + a_1, b_2 + a_2, \dots, b_n + a_n)$$

$$= (b_1, b_2, \dots, b_n) + (a_1, a_2, \dots, a_n)$$

$$= B + A$$

..

Properties of Vector Operations

Addition and Scalar Multiplication

$$1. \quad \vec{a} + \vec{b} = \vec{b} + \vec{a} \qquad 2. \quad \vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$$

$$3. \quad \vec{a} + \vec{0} = \vec{a} \qquad 4. \quad \vec{a} + (-\vec{a}) = \vec{0}$$

$$5. \quad c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b} \qquad 6. \quad (c + d)\vec{a} = c\vec{a} + d\vec{a}$$

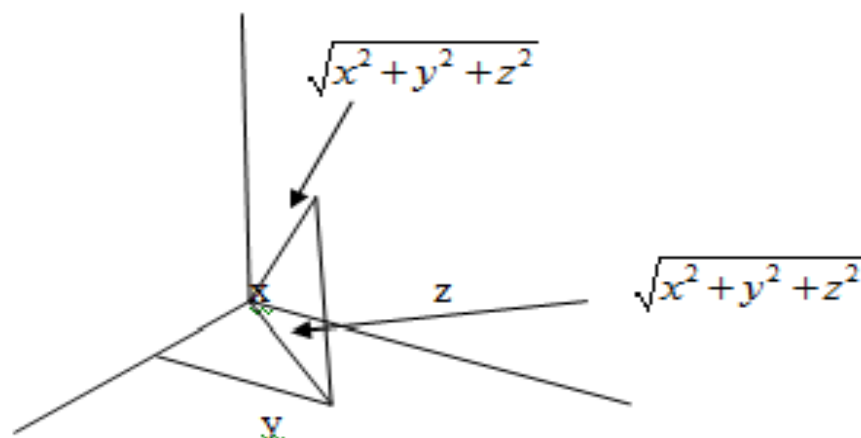
$$7. \quad (cd)\vec{a} = c(d\vec{a}) \qquad 8. \quad 1\vec{a} = \vec{a}$$

Exercise 2.2.1:

1. Given three vectors $A = (1, 1, 1)$, $B = (-1, 2, 3)$ and $C = (0, 3, 4)$, find
 - a. $A+B$
 - b. $2A - B$
 - c. $A+B - C$
 - d. $A - 3B + 10C$
2. Determine whether α and β can be found to satisfy the vector equations
 - a. $(2, 1, 0) = \alpha(-2, 0, 2) + \beta(1, 1, 1)$
 - b. $(-3, 1, 2) = \alpha(-2, 0, 2) + \beta(1,1,1)$

The Distance Formula

The distance formula is derived from the three dimensional version of the Pythagorean theorem, which is displayed below,



The distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2.3 Scalar product and norm of vector

Let $A = (a_1, a_2, \dots, a_n)$ and $B = (b_1, b_2, \dots, b_n)$ be two vectors. The **scalar product** of A and B is the number $A.B$ defined by

$$A.B = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

Note: The scalar product is also called a **dot product** or **inner product**.

Example 2.3.1

- 1) Given $A = (3, -1)$ and $B = (2, 3)$, then $A.B = 3$
- 2) $i.j = 0$, where $i = (1, 0, 0)$ and $j = (0, 1, 0)$.

Example 2.3.1

1) Given $A = (3, -1)$ and $B = (2, 3)$, then $A \cdot B = 3$

2) $\mathbf{i} \cdot \mathbf{j} = 0$, where $\mathbf{i} = (1, 0, 0)$ and $\mathbf{j} = (0, 1, 0)$.

The scalar product satisfies many of the laws that hold for real numbers. The basic ones are:

a) $A \cdot B = B \cdot A$

b) $t(A \cdot B) = (tA) \cdot B = A \cdot (tB)$

c) $(A + B) \cdot C = A \cdot C + B \cdot C$

d) If $A = \mathbf{0}$ is the zero vector, then $A \cdot A = 0$, and otherwise $A \cdot A > 0$

Activity 2.3.1: Proof the above properties.

Example 2.3.2: Given $A = (3, 2, -1)$ and $B = (2, 0, 3)$, and $C = (1, -1, 1)$, then

a. $A \cdot B = B \cdot A = 3$

b. $2(A \cdot B) = 6$

c. $(A+B) \cdot C = A \cdot C + B \cdot C = 5$

Activity 2.3.2: Find $A \cdot A$, $B \cdot B$, and $C \cdot C$. Are all positive values?

Dot Product

The dot product is defined by

$$\begin{aligned}\vec{a} &= \langle a_1, a_2, a_3 \rangle, \quad \vec{b} = \langle b_1, b_2, b_3 \rangle \\ \implies \vec{a} \cdot \vec{b} &= a_1 b_1 + a_2 b_2 + a_3 b_3\end{aligned}$$

and obeys

0. $\vec{a}, \vec{b}$ are vectors and $\vec{a} \cdot \vec{b}$ is a number
1. $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
2. $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
3. $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
4. $(c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b})$
5. $\vec{0} \cdot \vec{a} = 0$
6. $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$
7. $\vec{a} \cdot \vec{b} = 0 \iff \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0} \text{ or } \vec{a} \perp \vec{b}$

In property 6, θ is the angle between $\vec{a}$ and $\vec{b}$.

Exercise 2.3.1: Given the vectors $A = (1, -1, 2)$, $B = (-2, 0, 2)$, $C = (3, 2, 1)$. Evaluate

- a. $A \cdot C$ b. $B \cdot C$ c. $(A + B) \cdot C$ d. $A \cdot (2B + 3C)$ e. $(A \cdot B)C$

The **length**, or **norm** or **magnitude** of vector $A = (a_1, a_2, \dots, a_n)$, denoted by $\|A\|$, can be expressed in terms of the scalar product. By definition

$$\|A\|^2 = a_1^2 + a_2^2 + \dots + a_n^2$$

$$\text{and } A \cdot A = (a_1, a_2, \dots, a_n) \cdot (a_1, a_2, \dots, a_n) = a_1^2 + a_2^2 + \dots + a_n^2$$

Thus, the norm of the vector A is the number:

$$\|A\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$

Example 2.3.3: If $A = (-1, 2, 3)$ then $\|A\| = \sqrt{1+4+9} = \sqrt{14}$

Note: 1. A **unit vector** is a vector having norm or length 1.

$$2. \quad \|A\| = 0 \Leftrightarrow A = 0$$

Example 2.3.4: If $A = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$, then $\|A\| = \sqrt{(1/3) + (1/3) + (1/3)} = 1$

Note: Any non-zero vector can be fully represented by providing its magnitude and a unit vector along its direction. Let $\hat{A}$ be a **unit vector in the direction of A**. Then

$$\hat{A} = \frac{A}{\|A\|}$$

Example 2.3.5: Given a vector $A = (1, 1, 1)$. Find a unit vector in the direction of A.

Solution:

$\|A\| = \sqrt{1+1+1} = \sqrt{3}$, then the unit vector in the direction of A is:

$$\hat{A} = \frac{A}{\|A\|} = \frac{(1,1,1)}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) = \frac{1}{\sqrt{3}} (1,1,1)$$

Activity 2.3.3: 1. Given three vectors $A = (1, 1, 1)$, $B = (-1, 2, 3)$ and $C = (0, 3, 4)$, find the unit vector in the direction of $A+B - C$.

2. The vectors $\hat{i} = (1, 0, 0)$, $\hat{j} = (0, 1, 0)$ and $\hat{k} = (0, 0, 1)$ are unit vectors in the direction of positive x, y and z axis, respectively. Find a unit vector in the direction of $A = (-1, 2, 3)$.

Let A, B be two n -tuples of vectors. We define the distance between A and B to be

$$\|A - B\| = \sqrt{(A - B) \cdot (A - B)}$$

Note: Let A be any vector and $x \in \mathbb{R}$, then $\|A\| = \|-A\|$ and $\|xA\| = |x|\|A\|$

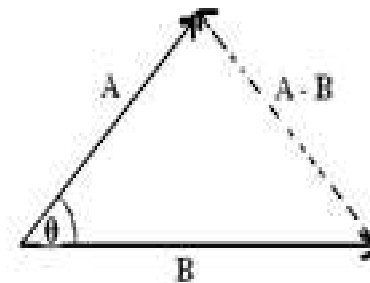
The following theorem gives us a geometric interpretation for the scalar product.

Theorem 2.3.1: Let $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$

be non-zero vectors and let θ be

the angle between A and B ($0 \leq \theta \leq \pi$).

Then $A \cdot B = \|A\| \|B\| \cos \theta$



Proof: Consider a triangle formed by $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A} - \mathbf{B}$

$$\|\mathbf{A} - \mathbf{B}\|^2 = \|\mathbf{A}\|^2 + \|\mathbf{B}\|^2 - 2\|\mathbf{A}\| \|\mathbf{B}\| \cos \theta \quad (\text{Why?})$$

$$\Rightarrow (a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 = a_1^2 + a_2^2 + a_3^2 + b_1^2 + b_2^2 + b_3^2 - 2\|\mathbf{A}\| \|\mathbf{B}\| \cos \theta$$

After cancellation, we get, $\mathbf{a}_1 \mathbf{b}_1 + \mathbf{a}_2 \mathbf{b}_2 + \mathbf{a}_3 \mathbf{b}_3 = \|\mathbf{A}\| \|\mathbf{B}\| \cos \theta$

$$\Rightarrow \mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \|\mathbf{B}\| \cos \theta$$

Activity 2.3.4: Given two non-zero vectors $\mathbf{A}$ and $\mathbf{B}$, how do you find the angle between them? Take, for example, $\mathbf{A} = (2, -1, 2)$, $\mathbf{B} = (1, -1, 0)$ and find the angle between them.

Note: Two non-zero vectors are said to be **orthogonal (Perpendicular)** if the angle between them is $\frac{\pi}{2}$.

Note: Two non-zero vectors A and B are said to be orthogonal (Perpendicular) if $A \cdot B = 0$.

Activity 2.3.5: 1. Which of the following pairs of vectors are perpendicular?

- $(1, -1, 1)$, and $(2, 1, 5)$
- $(-5, 2, 7)$, and $(3, -1, 2)$
- $(1, -1, 1)$, and $(2, 3, 1)$
- $(\pi, 2, 1)$, and $(2, -\pi, 0)$

2. Suppose $A \cdot B = A \cdot C$, what can you deduce about A , B and C ?

Exercise 2.3.2: 1. Show that $\|A - B\| = \|A + B\|$ if and only if $A \cdot B = 0$.

2. Let $A_1, A_2, \dots, A_r$ be non-zero vectors such that $A_i \cdot A_j = 0$ if $i \neq j$.

Let $c_1, c_2, \dots, c_r$ be numbers such that $c_1 A_1 + c_2 A_2 + \dots + c_r A_r = 0$.

Show that $c_i = 0$ for all $i = 1, 2, 3, \dots, r$.

Theorem 2.3.2 Let $\mathbf{A}$ and $\mathbf{B}$ be vectors. Then

a) $|\mathbf{A} \cdot \mathbf{B}| \leq \|\mathbf{A}\| \|\mathbf{B}\|$ (Schwarz inequality)

b) $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$ (Triangle inequality)

Proof a) If one of $\mathbf{A}$ or $\mathbf{B}$ is a zero vector then both sides of the inequality are equal to 0. Suppose both $\mathbf{A}$ and $\mathbf{B}$ are non-zero.

From $\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \|\mathbf{B}\| \cos \theta$,

$$|\mathbf{A} \cdot \mathbf{B}| = \|\mathbf{A}\| \|\mathbf{B}\| |\cos \theta| \text{ Which implies } |\mathbf{A} \cdot \mathbf{B}| \leq \|\mathbf{A}\| \|\mathbf{B}\| \quad (\text{Why?})$$

b) From the fact that $\|V\|^2 = V.V$,

$$\begin{aligned}\|A+B\|^2 &= (A+B) \cdot (A+B) \\ &= A.A + 2A.B + B.B \\ &= \|A\|^2 + 2A.B + \|B\|^2 \\ &\leq \|A\|^2 + 2|A.B| + \|B\|^2 && \text{(why ?)} \\ &\leq \|A\|^2 + 2\|A\| \|B\| + \|B\|^2 && \text{(why ?)} \\ &= (\|A\| + \|B\|)^2\end{aligned}$$

By taking the square roots,

$$\|A+B\| \leq \|A\| + \|B\|$$

Remark: The inequalities of Theorem 2.3.2 hold true also for any vectors A and B in $\mathbb{R}^n$.

Problem

1. Let $V = (-2, 3, 0, 6)$, find scalar K such that $\|KV\| = 5$

Solution: Given $V = (-2, 3, 0, 6)$ and $\|KV\| = 5$

Require k

$$\|v\| = \sqrt{(-2)^2 + 3^2 + 0^2 + 6^2} = 7$$

$$\text{but we know that } \|kv\| = |k|\|v\|$$

$$5 = |k|\|v\|$$

$$5 = |k|7$$

$$|k| = \frac{5}{7}$$

$$k = \frac{5}{7} \text{ or } k = -\frac{5}{7}$$

2. Find $U.V$ if $\|U + V\| = 1$ and $\|U - V\| = 5$

Solution Given $\|U + V\| = 1$ and $\|U - V\| = 5$

Required, $U.V$

$$\begin{aligned}\|u + v\|^2 &= (u + v).(u + v) = u.(u + v) + v.(u + v) \\ &= u.u + u.v + v.u + v.v \\ &= \|u\|^2 + 2u.v + \|v\|^2\end{aligned}$$

$$\text{Hence, } 1 = \|u\|^2 + 2u.v + \|v\|^2 \dots\dots\dots(1)$$

$$\text{Similarly, } \|u - v\|^2 = (u - v).(u - v) = \|u\|^2 - 2u.v + \|v\|^2$$

$$\text{Hence } 5 = \|u\|^2 - 2u.v + \|v\|^2 \dots\dots\dots(2)$$

Subtract equation (1) from (2), then we get

$$5 - 1 = -4u.v$$

$$u.v = -1$$

Hence the result

3. Given that $\|V\| = 3$ and $\|W\| = 5$ find the value of x for which the vectors $V+xW$ and $V-xW$ will be orthogonal.

Solution: Given $\|V\| = 3$ and $\|W\| = 5$ the vectors $V+xW$ and $V-xW$ will be orthogonal.

Required: find x

$V+xW$ and $V-xW$ are orthogonal means

$$(V + xW) \cdot (V - xW) = 0$$

$$V \cdot (V - xW) + xW \cdot (V - xW) = 0$$

$$V \cdot V - xV \cdot W + xW \cdot V - xW \cdot xW = 0$$

$$\|V\|^2 - x^2\|W\|^2 = 0$$

$$3^2 - x^2 5^2 = 0$$

$$25x^2 = 9$$

$$x^2 = \frac{9}{25}$$

$$\text{there fore } x = \frac{3}{5} \text{ and } x = -\frac{3}{5}$$

2.4 The Vector product

The second type of product of two vectors is the cross product. Unlike the dot product, the cross product of two vectors is a vector.

Definition 1.4.1 The cross product (or vector product) $A \times B$ of two vectors

Let $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$ is defined by

$$A \times B = \det \begin{pmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}$$

$$A \times B = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

Note that the cross product is defined in $\mathfrak{R}^3$

Example 2.4.1 Let $A = (2, -1, 3)$ and $B = (-1, -2, 4)$

$$A \times B = (-4 + 6, -3 - 8, -4 - 1) = (2, -11, -5)$$

Activity 2.4.1: Find $B \times A$.

Theorem 2.4.1 For vectors A , B and C ,

$$1) A \times B = -(B \times A)$$

$$2) A \times A = O$$

$$3) tA \times B = t(A \times B) = A \times (tB), \quad t \in \mathbb{R}$$

$$4) \|A \times B\|^2 = \|A\|^2 \|B\|^2 - (A \cdot B)^2$$

$$5) C \cdot (A \times B) = B \cdot (C \times A) = A \cdot (B \times C)$$

$$6) (A + B) \times C = (A \times C) + (B \times C)$$

$$7) C \times (A + B) = (C \times A) + (C \times B)$$

$$8) A \cdot (A \times B) = 0 \text{ and } B \cdot (A \times B) = 0 \text{ (that is, } A \times B \text{ is perpendicular to both } A \text{ and } B.)$$

$$9) (A \times B) \times C = (A \cdot C) B - (B \cdot C) A$$

Proof :The following is the proof for 1, 2 and 8. The rest are left as an exercise

1) From the definition of cross product,

$$\underline{A \times B} = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

For $\underline{B \times A}$, interchange A and B to obtain

$$\begin{aligned}\underline{B \times A} &= (b_2 a_3 - b_3 a_2, b_3 a_1 - b_1 a_3, b_1 a_2 - b_2 a_1) \\ &= (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1) \\ &= -(\underline{A \times B})\end{aligned}$$

$$\begin{aligned}2) \quad \underline{A \times A} &= (a_2 a_3 - a_3 a_2, a_3 a_1 - a_1 a_3, a_1 a_2 - a_2 a_1) \\ &= (0, 0, 0)\end{aligned}$$

8) Setting $\underline{C} = \underline{A}$ in 5) yields

$$\begin{aligned}\underline{A} \cdot (\underline{A \times B}) &= \underline{B} \cdot (\underline{A \times A}) \\ &= \underline{B} \cdot \underline{0} \text{ (why?)} \\ &= 0\end{aligned}$$

By setting $\underline{C} = \underline{B}$ in 5),

$$\begin{aligned}\underline{B} \cdot (\underline{A \times B}) &= \underline{A} \cdot (\underline{B \times B}) \\ &= \underline{A} \cdot \underline{0} = 0\end{aligned}$$

This shows that for non zero vectors A and B, the cross product $\underline{A \times B}$ is orthogonal to both A and B.

Activity 2.4.2: Are the usual commutative and associative laws valid? i.e. for any vectors A, B and C in $\mathbb{R}^3$, is $A \times B = B \times A$?
Is $A \times (B \times C) = (A \times B) \times C$?

Exercise 2.4.1: Let $A = (2, 1, 0)$, $B = (2, -1, 1)$ and $C = (0, 1, 1)$. Find

- a. $A \times B$ b. $(A \times B) \times C$ c. $(A \cdot C)B - (B \cdot C)A$ d. $B \times C$ e. $A \times (B \times C)$

From 4) of theorem 1.4.1, we derive an important formula for the norm of the cross product.

$$\begin{aligned}\|A \times B\|^2 &= \|A\|^2 \|B\|^2 - (A \cdot B)^2 \\ &= \|A\|^2 \|B\|^2 - \|A\|^2 \|B\|^2 \cos^2 \theta \quad (\theta \text{ is the angle between } A \text{ and } B) \\ &= \|A\|^2 \|B\|^2 (1 - \cos^2 \theta) \\ &= \|A\|^2 \|B\|^2 \sin^2 \theta \\ \Rightarrow \|A \times B\| &= \|A\| \|B\| \sin \theta \quad (\text{For } 0 \leq \theta \leq \pi, \sin \theta \text{ is non-negative})\end{aligned}$$

Activity 2.4.3:

- For the unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, find $\mathbf{i} \times \mathbf{j}$, $\mathbf{j} \times \mathbf{k}$ and $\mathbf{k} \times \mathbf{i}$. What is $\mathbf{j} \times \mathbf{i}$?
- If $\mathbf{A}$ and $\mathbf{B}$ are parallel, what is $\mathbf{A} \times \mathbf{B}$?
- If $\mathbf{A}$ and $\mathbf{B}$ are orthogonal, What is $\|\mathbf{A} \times \mathbf{B}\|$?

Exercise 2.4.1:

1. Find a unit vector perpendicular to both $\mathbf{A} = (2, -3, 1)$ and $\mathbf{B} = (1, 2, -4)$.
2. Prove that $(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} + \mathbf{B}) = 2(\mathbf{A} \times \mathbf{B})$.

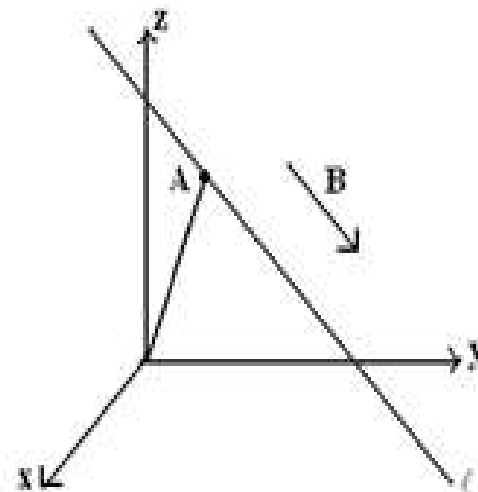
2.6 Lines and planes

Definition 2.6.1: A line ℓ is any set of the form $\{\mathbf{p} : \mathbf{p} = \mathbf{A} + t\mathbf{B}, t \in \mathbb{R}\}$ where $\mathbf{B}$ is assumed to be a non-zero vector and $\mathbf{A}$ is a fixed point on the line.

Note that if (x, y, z) is on line ℓ and if $\mathbf{A} = (a_1, a_2, a_3)$ and $\mathbf{B} = (b_1, b_2, b_3)$ then $(x, y, z) = (a_1, a_2, a_3) + t(b_1, b_2, b_3)$ for some real number t .

Activity 2.6.2: Is point $\mathbf{A}$ on ℓ ? Is $\mathbf{B}$ parallel to a vector formed by any two points of ℓ ?

$\mathbf{P} = \mathbf{A} + t\mathbf{B}$ is a vector equation of a line through $\mathbf{A}$



Exercise 2.6.1: Let the line L_1 passes through the points $(5, 1, 7)$ and $(6, 0, 8)$ and the line L_2 passes through the points $(3, 1, 3)$ and $(-1, 3, \alpha)$. Find the value of α for which the two lines intersect.

Suppose $P = (x, y, z)$ is a point on line ℓ through $A = (a_1, a_2, a_3)$ in the direction of $B = (b_1, b_2, b_3)$. Then $p = A + tB \Rightarrow (x, y, z) = (a_1, a_2, a_3) + t(b_1, b_2, b_3)$ or equivalently

$$x = a_1 + b_1 t$$

$$y = a_2 + b_2 t$$

$$z = a_3 + b_3 t$$

These equations are **parametric equation** of a line and t is called a **parameter**.

Activity 2.6.3: 1) Find the parametric equation of a line that contains $(2, -1, 1)$ and is parallel to the vector $(3, \frac{1}{2}, 0)$.

2) From the parametric equation of a line in $\mathbb{R}^3$, derive the equation

$$\frac{y - a_1}{b_1} = \frac{y - a_2}{b_2} = \frac{z - a_3}{b_3}$$

It is called **standard form** of equation of a line.

If the line is on a plane show that the standard form reduces to an equation of the form $y = mx + c$.

2.6. Equations of Lines & Planes

Definition

A Line in the space is determined by **a point and a direction**.

Consider a line L and a point $p(x_0, y_0, z_0)$ on L.

Direction of this line is determined by a vector v that **is parallel to Line L**.

Let $P(x, y, z)$ be any point on the Line

Let $\vec{r}_0$ is the Position vector of point p_0 , $\vec{r}$ is the Position vector of point P

Then **vector equation of line** is given by

$$\vec{r} = \vec{r}_0 + vt \quad \text{Where } t \text{ is a scalar}$$

Let $v = (a, b, c)$ $v = (a, b, c)$ $r = (x_0, y_0, z_0)$

Hence the **parametric equation of a line** is:-

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$$

$$(x, y, z) = (x_0 + at, y_0 + bt, z_0 + ct)$$

Hence we get,

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

• Symmetric Equations of line is given by:-

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Example 1

Find the vector equation and

Parametric equations of the line through

$(1,2,3)$ and parallel to vector $3i+2j-k$

Solution:- Given $v= 3i+2j-k$

$$\vec{r_0} = (1,2,3) = i+2j+3k$$

Hence the **vector equation of line** is

$$r = i+2j+3k+t(3i+2j-k)$$

$$= (1+3t)\underline{i} + (2+2t)\underline{j} + (3-t)\underline{k}$$

$$\text{Since } r = x\underline{i} + y\underline{j} + z\underline{k}$$

Hence we get **parametric equation of line** is:-

$$x = 1+3t$$

$$y = 2+2t$$

$$z = 3-t$$

Example2

Find the symmetric equation for line through point(1,-5,6) and is parallel to vector (-1,2,-3)

Solution:-We know that symmetric equation of line is given by:-

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}, \text{ here } a=-1, b=2, c=-3$$

Hence we get

$$\frac{x-1}{-1} = \frac{y+5}{2} = \frac{z-6}{-3}$$

Hence the result

Example3: Find the parametric and symmetric equations of the line through $P_1 (1; 2; 3)$ and $P_2 (2; 4; 1)$.

Solution: First, we need to find the direction vector. Since the line goes through P_1 and P_2 , the vector $\overrightarrow{p_1 p_2}$ will be parallel to the line.

$\overrightarrow{p_1 p_2} = (1, 2, -2)$, We get

$$x = 1 + t$$

$$y = 2 + 2t$$

$$z = 3 - 2t$$

If we solve for t , we get $x - 1 = \frac{y - 2}{2} = \frac{z - 3}{-2}$

Example 4. Let L_1 be the line through $(1; 6; 2)$ with direction vector $(1; 2; 1)$ and L_2 be the line through $(0; 4; 1)$ with direction vector $(2; 1; 2)$. Determine if the lines are parallel, if they intersect or if they are skew. If they intersect, find the point at which they intersect.

solution : For the lines to be parallel, their direction vectors would have to be parallel, that is there would have to exist a constant c such that

$$(1; 2; 1) = c (2; 1; 2)$$

Which would imply

$$1 = 2c$$

$$2 = c$$

$$1 = 2c$$

Which has no solution. So, the lines are not parallel.

We now derive the equation of each line.

– For $L1$

$$x = 1 + t$$

$$y = -6 + 2t$$

$$z = 2 + t$$

– For $L2$

$$x = 2s$$

$$y = 4 + s$$

$$z = 1 + 2s$$

The line will intersect if we can solve

$$1 + t = 2s$$

$$-6 + 2t = 4 + s$$

$$2 + t = 1 + 2s$$

Equation 1 gives

$$t = 2s - 1$$

If we substitute in equation 2, we get

$$6 + 2t = 4 + s$$

$$6 + 2(2s - 1) = 4 + s$$

$$6 + 4s - 2 = 4 + s$$

$$3s = 12$$

$$s = 4$$

Since we had $t = 2s - 1$ this implies that $t = 7$. We need to verify that these values also work in equation 3. Replacing s and t by their values gives us

$$2 + 7 = 1 + 2(4)$$

$$9 = 9$$

so, the two lines intersect. To find the point of intersection, we can use the equation of either line with the value of the corresponding parameter. If we use L_1 , the parameter is t and the value of t which gives the point of intersection is $t = 7$. Thus, the point of intersection is

$$x = 1 + 7$$

$$y = 6 + 2(7)$$

$$z = 2 + 7$$

That is $x = 8$, $y = 8$, $z = 9$

vvvvvv

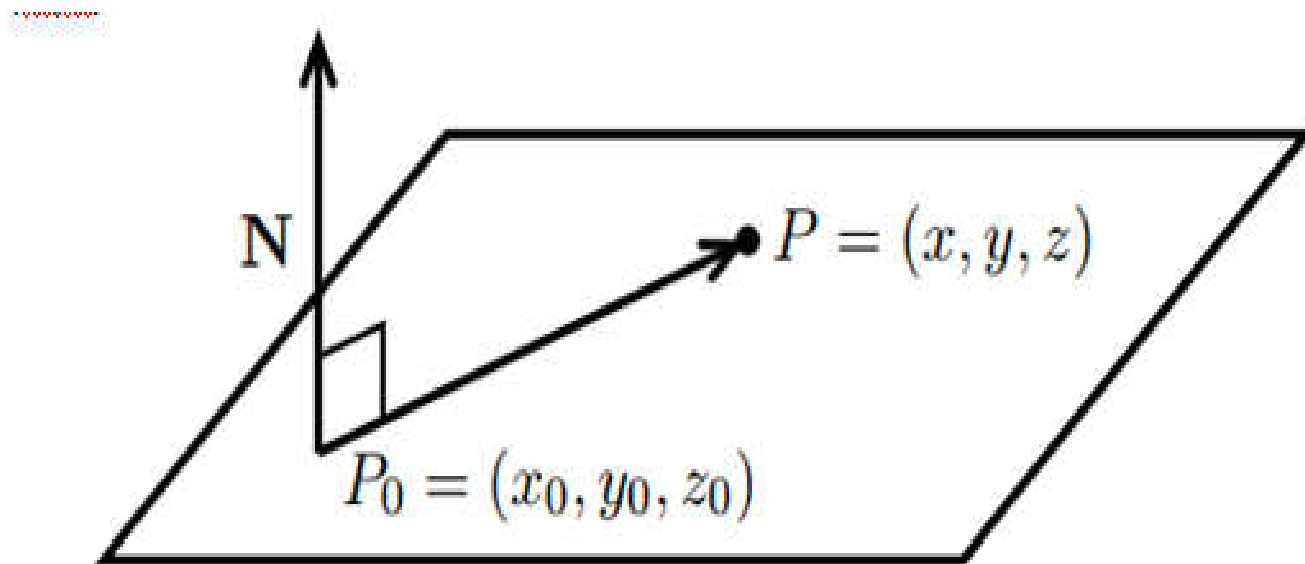
Example 5. Find the angle between the two previous lines.

Their direction vectors are $u = (1; 2; 1)$ and $v = (2; 1; 2)$. The smallest angle between these vectors is

$$\theta = \cos^{-1} \frac{u \cdot v}{\|u\| \|v\|} = \cos^{-1} \frac{2+2+2}{3\sqrt{9}} = 35.26^\circ$$

Planes

- The plane in the space is determined by a point and a vector that is perpendicular to plane.
- Let $p(x_0, y_0, z_0)$ be given point and N is the orthogonal vector.
- Let $P(x, y, z)$ be any point in space and r, r_0 is the position vector of point P and p_0 respectively.
- Then **vector equation of plane** is given by:-
$$Nr = Nr_0$$
- Let $N = (a, b, c)$
- $r = (x, y, z)$



- $r_0 = (x_0, y_0, z_0)$
- Hence the **vector equation** becomes:-
- $(a, b, c) \cdot (x - x_0, y - y_0, z - z_0) = 0$
- Hence the **scalar equation of plane** is given by
- $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

- We can also write above equation of plane as:-
- $ax+by+cz+d=0$
- Where $d = -(ax_0 + by_0 + cz_0)$

Example 3

Find the equation of plane through point $(1,-1,1)$ and with normal vector $\mathbf{i}+\mathbf{j}-\mathbf{k}$

Solution:- Given point is $(1,-1,1)$

Here $a=1, b=1, c=-1$

We know that equation of plane is given by:-

$$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$

$$1(x-1)+1(y+1)-1(z-1)=0$$

$$x+y-z+1=0$$

Example 4

Find the line of intersection of two planes $x+y+z=1$ and $x+2y+2z=1$

• **Solution:-** Let L is the line of intersection of two planes.

We can find the point where Line L intersects

xy plane by setting $z=0$ in above two equations, we get:-

$$x+y=1$$

$$x+2y=1$$

- By solving for x and y we get,
- $x=1$
- $y=0$
- Hence the Point on Line L is $(1,0,0)$
- Line L lies in both planes so it is perpendicular to both normal vector
- $a=i+j+k$ and $b=i+2j+2k$
- Hence vector v which is given by
- $v=a \times b$ is parallel to L

- Hence
- $v=0i-j+k$
- We know that equation of line is given by

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

Hence we get,

- $x=1, y=-t, z=2t$

Example 5

Find the equation of plane through the points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$

Solution:-Let $p(0,1,1)$, $q(1,0,1)$ and $r(1,1,0)$ denotes the given points.

The goal is to find the basic data, i.e. a point in the plane and a normal to the plane. The point is easy; we already have three of them. To get the normal we note that $\overrightarrow{pq}$ and $\overrightarrow{pr}$ are vectors in the plane, so their cross product is orthogonal (normal) to the plane. That is,

Let $\vec{a} = \overrightarrow{pq} = (1-0, 0-1, 1-1) = (1, -1, 0)$ and

$$\vec{b} = \overrightarrow{pr} = (1-1, 1-0, 0-1) = (0, 1, -1)$$

$\vec{N} = \vec{a} \times \vec{b}$ is the orthogonal vector of the plane.

Hence $\vec{N} = \vec{i} + \vec{j} + \vec{k}$

• Hence the equation of plane through $(0, 1, 1)$ and perpendicular to vector $\vec{i} + \vec{j} + \vec{k}$ is:-

• $1(x-0) + 1(y-1) + 1(z-1) = 0$

• $x + y - 1 + z - 1 = 0$

• $x + y + z = 2$

• Hence the result

Practice Problems

- 1) Find the parametric equation of line through point $(1, -1, 1)$ and parallel to line $x+2=y/2=z-3$.
- 2) Find the equation of plane through points $(3, -1, 2)$, $(8, 2, 4)$ and $(-1, -2, -3)$
- 3) Find the symmetric equation of line of intersection of planes $5x-2y-2z=1$,
 $4x+y+z=6$
- 4) Find the point at which line $x=3-t, y=2+t, z=5t$ intersects the plane $x-y+2z=9$